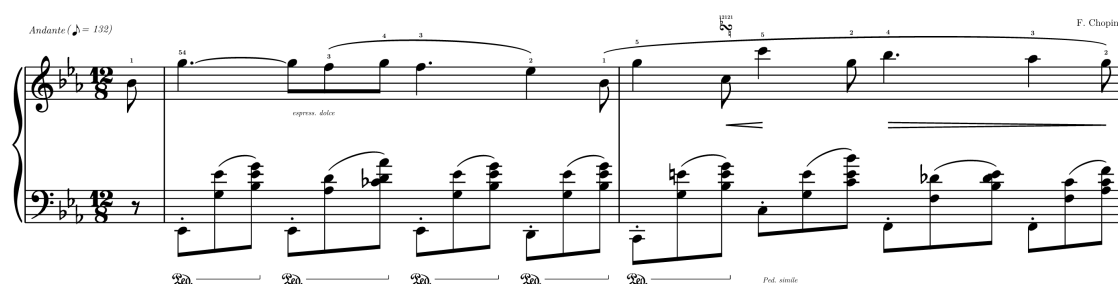


typed-scores

Western music notation, natively in Typst

User Guide



Version 0.5.1

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1 Introduction

typed-scores turns compact musical text into engraved western notation inside Typst. A Rust/WASM plugin parses pitches, durations, exact onsets, and beam groups; Typst and CeTZ lay out the score with bundled Bravura glyphs.

The package is designed for lecture notes, theory worksheets, analytical examples, short compositions, and reusable excerpts whose musical content should remain readable and versionable as text.

Note. The public API intentionally has only two entry points: `score` for complete scores and `bar` for a quick one-staff measure.

```
1 #score(  
2   time: "4/4",  
3   bars: ((notes: "c4:q d e f"),),  
4 )
```

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2 Getting started

2.1 Installation and import

Import the package from the Typst preview namespace:

```
1 #import "@preview/typed-scores:0.5.1": *
```

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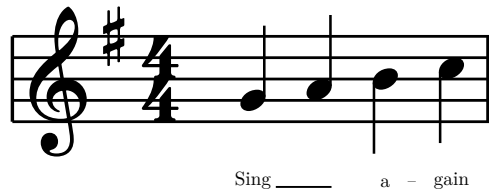
The examples in this guide use lowercase pitch letters for fast entry. Uppercase remains accepted with identical pitch and octave semantics.

2.2 Quick measure with `bar()`

`bar` is a convenience wrapper over the same score pipeline used by `score`. It accepts a note string, optional lyrics, and its clef, key, and optional meter.

```
1 #bar(  
2   "g4:q a b c5",  
3   lyrics: "Sing __ a -- gain",  
4   clef: "treble",  
5   key: "G",  
6   time: "4/4",  
7 )
```

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With `time` omitted, the `bar` remains useful for a short fragment and does not perform full-measure duration validation.

Argument	Type	Default	Description
<code>notes</code>	<code>str</code>	required	One event string.
<code>lyrics</code>	<code>str</code> <code>array</code> <code>none</code>	<code>none</code>	One lyric verse or an array of verses.
<code>clef</code>	<code>str</code>	<code>"treble"</code>	Treble, bass, alto, or tenor clef.
<code>key</code>	<code>str</code>	<code>"C"</code>	Major or minor key signature.
<code>time</code>	<code>str</code> <code>none</code>	<code>none</code>	Meter and expected duration, such as <code>"4/4"</code> .
<code>theme</code>	<code>str</code>	<code>"auto"</code>	Notation ink: <code>light</code> uses black, <code>dark</code> uses white, and <code>auto</code> follows the surrounding solid text fill.
<code>lyric-size</code>	<code>number</code>	<code>0.9</code>	Lyric text size in staff spaces.
<code>lyric-font</code>	<code>str</code> <code>none</code>	<code>none</code>	Optional font family for lyrics.
<code>lyric-gap</code>	<code>number</code>	<code>0.8</code>	Clearance above the first verse.
<code>verse-gap</code>	<code>number</code>	<code>1.45</code>	Baseline distance between verses; must be at least <code>lyric-size</code> .

2.3 Light and dark notation

Use `theme: "dark"` when placing a score on a dark surface. It recolors staff geometry, text, and every bundled Bravura glyph together. The default `theme: "auto"` follows the surrounding Typst text fill; it cannot inspect CSS changes applied by an HTML host after Typst has exported the SVG.

```
1 #block(fill: rgb("#202124"),  
  inset: 10pt)[  
2   #bar(  
3     "c4:q d e f",  
4     time: "4/4",  
5     theme: "dark",  
6   )  
7 ]
```

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3 Scores and systems

Every score is an ordered **bars**: list. Each bar is a dictionary: it contains staff event strings and optional metadata that belongs to that musical bar. Never place structural bar delimiters inside an event string.

3.1 One staff

For one staff, omit **staves**: and use the reserved **notes**: field in every bar. There is no artificial staff name to repeat. A string is one rhythmic voice; an array of two to four strings creates independent simultaneous voices on the same staff.

```
1 #score(  
2   clef: "treble",  
3   time: "4/4",  
4   bars: (  
5     (notes: "c5:q d5:q e5:q f5:q"),  
6     (notes: "g5:h e5:h"),  
7   ),  
8 )
```

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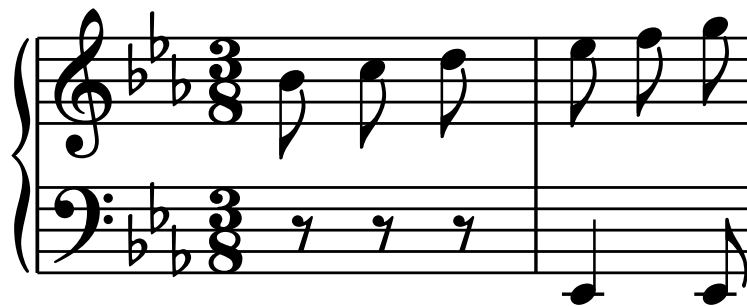
3.2 Multiple staves

For two or more staves, declare a map of permanent staff IDs and clefs. Every bar must then contain exactly one string or voice-string array for each declared ID.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    key: "Eb",
7    time: "3/8",
8    bars: (
9      (
10       upper: "bb4:e c5:e d5:e",
11       lower: "r:e r:e r:e",
12     ),
13     (
14       upper: "eb5:e f5:e g5:e",
15       lower: "eb2:q eb2:e",
16     ),
17   ),
18   beams: true,
19 )

```

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Staff IDs are Typst dictionary keys, so keep them short and machine-oriented. They cannot reuse reserved bar fields such as `notes`, `lyrics`, `key`, `time`, `partial`, `tempo`, or `harmony`.

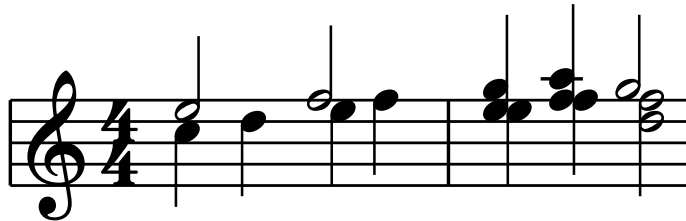
3.3 Independent voices

Use an array when notes on one staff have independent rhythms. The voice count must stay constant for that staff throughout the score. Following LilyPond's standard polyphonic convention, the first voice uses upward stems, the second downward stems, and later voices alternate. Rests move into separate vertical lanes. Coincident identical noteheads merge; unisons and seconds that cannot merge are shifted horizontally while retaining their shared onset.

```

1  #score(
2    time: "4/4",
3    bars: (
4      (notes: (
5        "e5:h f5:h",
6        "c5:q d5:q e5:q f5:q",
7      )),
8      (notes: (
9        "(e5 g5):q (f5 a5):q g5:h",
10       "e5:q f5:q (d5 f5):h",
11      )),
12    ),
13  )

```

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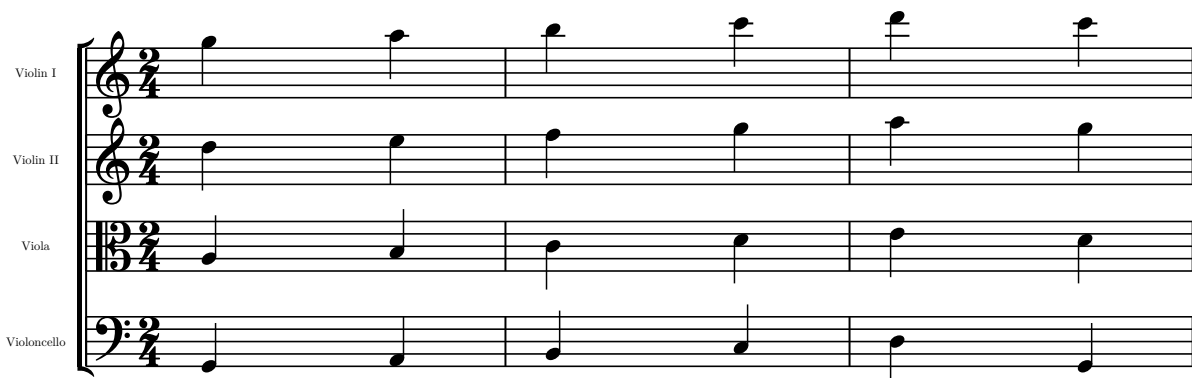
Each staff may also define `label` for its first-system name and `short-label` for the optional name on later systems. This follows LilyPond's `instrumentName` / `shortInstrumentName` behavior: names live in a shared column to the left of the group, center horizontally within that column, and center vertically on their individual staff. The compact example below shows the full first-system names; wrapped systems substitute the short names.

```

1  #score(
2      staves: (
3          violin-one: (clef: "treble", label: "Violin I", short-label: "Vln. I"),
4          violin-two: (clef: "treble", label: "Violin II", short-label: "Vln. II"),
5          viola: (clef: "alto", label: "Viola", short-label: "Vla."),
6          cello: (clef: "bass", label: "Violoncello", short-label: "Vc."),
7      ),
8      group: "bracket",
9      time: "2/4",
10     scale: 0.58,
11     ragged-right: false,
12     bars: (
13         (violin-one: "g5:q a5:q", violin-two: "d5:q e5:q", viola: "a3:q b3:q", cello:
14             "g2:q a2:q"),
15         (violin-one: "b5:q c6:q", violin-two: "f5:q g5:q", viola: "c4:q d4:q", cello:
16             "b2:q c3:q"),
17         (violin-one: "d6:q c6:q", violin-two: "a5:q g5:q", viola: "e4:q d4:q", cello:
18             "d3:q g2:q"),
19     ),
20 )

```

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The `group` argument controls the symbol at the left of a multi-staff system: `"brace"`, `"bracket"`, `"line"`, or `"none"`. Its default, `auto`, uses no symbol for one staff, a brace for two staves, and a bracket for larger ensembles. Brace groups connect barlines across their staves; the other styles keep each staff's barline separate.

3.4 Bar metadata

`clef`, `key`, `time`, and `tempo` may be set on any bar and persist until changed. `partial` changes duration validation only: it does not change the displayed meter. When a key change removes old alterations, cancellation naturals are engraved before the replacement key signature. This replaces separate section or parallel-array forms.

For a single staff, a clef change is `clef: "bass"`. For multiple staves, use a dictionary containing only the staves that change, such as `clef: (left: "bass")`. As in LilyPond, a mid-system change is reduced in size; the active clef is printed full-size when a new system begins.

```

1  #score(
2      clef: "treble",
3      key: "C",
4      time: "4/4",
5      bars: (
6          (
7              notes: "c5:q d5:q e5:q f5:q",
8              tempo: (text: [Moderato], beat: "quarter", bpm: 108),
9          ),
10         (
11             key: "Eb",
12             time: "3/4",
13             notes: "eb5:q f5:q g5:q",
14         ),
15     ),
16 )

```

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Moderato (♩ = 108)



For a metronome mark, use a tempo dictionary rather than inserting a Unicode note character. `beat` accepts `whole`, `half`, `quarter`, `eighth`, `sixteenth`, or `thirty-second`, and the matching note glyph is drawn automatically. `text` is optional, so `(beat: "eighth", bpm: 132)` produces just the note-and-number mark.

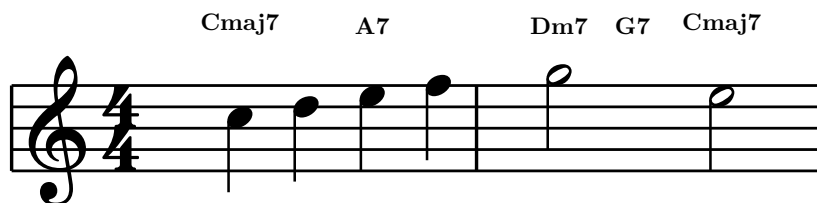
3.5 Harmony symbols

Use a bar's `harmony` field for chord symbols above the top staff. It is an independent duration-bearing sequence: each token is author-controlled text followed by `:` and a duration, and the sequence must fill the active bar. Harmony changes use the same onset positions as the notes, including changes that fall between melody note onsets. Each symbol is centered on the onset where the harmony becomes active, independent of its duration. Symbols are text, so jazz spellings such as `Cmaj7`, `F#7(b9)`, `Bb/D`, and `N.C.` remain under the author's control.

```

1  #score(
2    time: "4/4",
3    bars: (
4      (
5        notes: "c5:q d5:q e5:q f5:q",
6        harmony: "Cmaj7:h A7:h",
7      ),
8      (
9        notes: "g5:h e5:h",
10       harmony: "Dm7:q G7:q Cmaj7:h",
11      ),
12    ),
13  )

```

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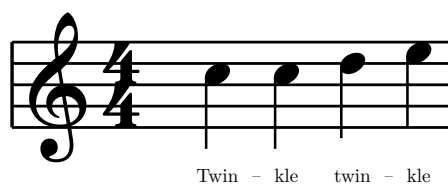
3.6 Lyrics

Use a bar's `lyrics` field to align syllables with the main pitched events of a staff's first voice. A plain string creates one verse. `--` draws a hyphen without consuming an event, `_` consumes one event and extends the preceding syllable, and `_` consumes one event without printing text. Written rests and grace notes do not consume lyric tokens. Every verse present in a bar must account for all of that bar's main pitched events, which keeps alignment mistakes visible instead of silently shifting later words.

```

1  #score(
2    time: "4/4",
3    bars: (
4      (
5        notes: "c5:q c d e",
6        lyrics: "Twin -- kle twin -- kle",
7      ),
8    ),
9  )

```

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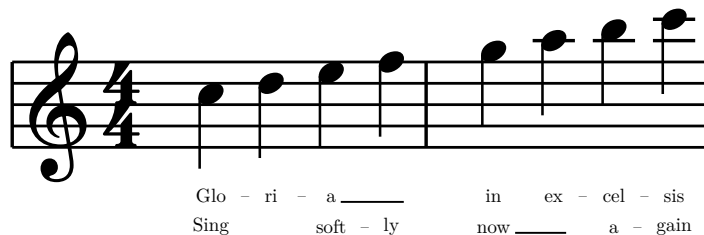
An array creates multiple verses. Use an underscore inside a syllable token to print a space, while a standalone underscore remains a skip. Hyphens and melisma extenders may cross barlines and resume when the score wraps.

```

1  #score(
2    time: "4/4",
3    width: 34,
4    bars: (
5      (
6        notes: "c5:q d e f",
7        lyrics: ("Glo -- ri -- a __", "Sing _ soft -- ly"),
8      ),
9      (
10       notes: "g5:q a b c6",
11       lyrics: ("in ex -- cel -- sis", "now __ a -- gain"),
12     ),
13   ),
14 )

```

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In a declared multi-staff score, `lyrics` is a dictionary whose keys are staff IDs. Each value is one verse string or an array of verses. Omit staves that have no lyrics in that bar. Lyrics always follow the first voice of their target staff; use that voice for the sung line when a staff is polyphonic.

```

1  lyrics: (
2    soprano: ("Hal -- le -- lu -- jah", "Praise _ the Lord"),
3    bass: "Low __",
4  )

```

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Lyric widths constrain the shared onset grid before measures are packed, so long syllables can widen bars and change line breaks. Verse lanes are also part of the staff's vertical ink extent: they clear dynamics and other below-staff marks, enlarge automatic multi-staff gaps, and contribute to system bounds. Use `lyric-size`, `lyric-font`, `lyric-gap`, and `verse-gap` to tune their typography and vertical rhythm.

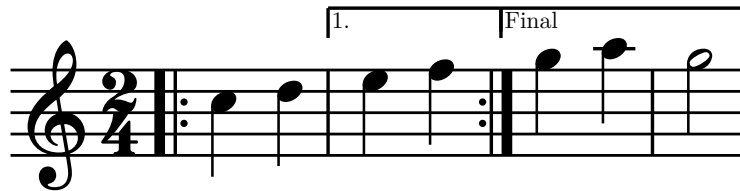
3.7 Barlines, navigation, and labeled endings

Repeat signs belong to a bar edge. Use `repeat-start` on a left edge and `repeat-end` on a right edge. A labeled ending is a span: mark its first bar with `start: true` and its last bar with `stop: true`. `label` is rendered literally, so punctuation and text such as "1." or "Final" are both valid.

```

1  #score(
2    time: "2/4",
3    bars: (
4      (barline: (left: "repeat-start"), notes: "c5:q d5:q"),
5      (
6        barline: (right: "repeat-end"),
7        ending: (label: "1.", start: true, stop: true),
8        notes: "e5:q f5:q",
9      ),
10     (ending: (label: "Final", start: true), notes: "g5:q a5:q"),
11     (ending: (label: "Final", stop: true), notes: "g5:h"),
12   ),
13 )

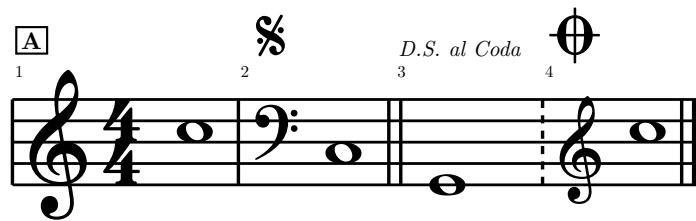
```

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A repeat end followed immediately by a repeat start becomes one double-sided repeat barline. Volta brackets continue across system wraps. Endings are currently sequential: one bracket must stop before another one starts.

A right bar edge may also be **double**, **final**, or **dashed**. Use a bar's `rehearsal` field for a boxed rehearsal label and `navigation` for "segno", "coda", arbitrary text, or Typst content. Marks attach to the beginning of their bar. When a bar number and mark coincide, the number stays closest to the staff and the mark stacks above it. Long boundary text reserves enough horizontal space to clear the following bar. `bar-numbers: "systems"` prints the first bar number of later systems, matching LilyPond's usual sparse numbering; `"all"` prints every bar number.

```
1  #score(  
2    time: "4/4",  
3    bar-numbers: "all",  
4    bars: (  
5      (rehearsal: "A", notes: "c5:w"),  
6      (clef: "bass", navigation: "segno", barline: (right: "double"), notes: "c3:w"),  
7      (navigation: "D.S. al Coda", barline: (right: "dashed"), notes: "g2:w"),  
8      (clef: "treble", navigation: "coda", barline: (right: "final"), notes: "c5:w"),  
9    ),  
10  )
```

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3.8 Score arguments

Argument	Type	Default	Description
<code>clef</code>	<code>str</code>	<code>"treble"</code>	Clef for the implicit single staff only.
<code>staves</code>	<code>dictionary</code> <code>none</code>	<code>none</code>	Permanent staff-ID map. Each spec has <code>clef</code> , plus optional first-system <code>label</code> and later-system <code>short-label</code> .
<code>bars</code>	<code>array</code>	<code>()</code>	Required non-empty array of bar dictionaries.
<code>key</code>	<code>str</code>	<code>"C"</code>	Initial key signature.
<code>time</code>	<code>str</code>	<code>"4/4"</code>	Initial meter and full-bar duration.
<code>theme</code>	<code>str</code>	<code>"auto"</code>	Notation ink: <code>light</code> uses black, <code>dark</code> uses white, and <code>auto</code> follows the surrounding solid text fill.
<code>tempo</code>	<code>str</code> <code>content</code> <code>dictionary</code> <code>none</code>	<code>none</code>	Tempo text, or <code>(text:, beat:, bpm:)</code> for an engraved metronome mark.
<code>composer</code>	<code>str</code> <code>content</code> <code>none</code>	<code>none</code>	Composer credit above the first system.
<code>width</code>	<code>number</code> <code>none</code>	<code>none</code>	Packing width in staff spaces.
<code>scale</code>	<code>number</code>	<code>1.0</code>	Uniform staff-space scale.
<code>note-spacing</code>	<code>number</code>	<code>3.1</code>	Horizontal density: the width in staff spaces of a quarter note in a bar of quarters; other durations scale logarithmically from the bar's shortest note.
<code>lyric-size</code>	<code>number</code>	<code>0.9</code>	Lyric text size in staff spaces.
<code>lyric-font</code>	<code>str</code> <code>none</code>	<code>none</code>	Optional lyric font family.
<code>lyric-gap</code>	<code>number</code>	<code>0.8</code>	Clearance between existing below-staff ink and the first verse.
<code>verse-gap</code>	<code>number</code>	<code>1.45</code>	Baseline distance between lyric verses; must be at least <code>lyric-size</code> .
<code>beams</code>	<code>bool</code>	<code>false</code>	Connect automatic beam groups.
<code>staff-gap</code>	<code>number</code> <code>none</code>	<code>none</code>	Fixed distance between adjacent staff lines.
<code>group</code>	<code>auto</code> <code>str</code>	<code>auto</code>	Auto, brace, bracket, line, or none.
<code>wrap</code>	<code>bool</code>	<code>true</code>	Wrap complete bars into systems.
<code>indent</code>	<code>number</code>	<code>0</code>	First-system indentation in staff spaces. The system still ends at the normal right edge.
<code>short-indent</code>	<code>number</code>	<code>0</code>	Indentation for every system after the first.

<code>ragged-right</code>	<code>auto bool</code>	<code>auto</code>	Leave every system at natural width. Auto is ragged only for a one-system score.
<code>ragged-last</code>	<code>bool</code>	<code>false</code>	Leave only the final system at its natural width.
<code>system-gap</code>	<code>length</code>	<code>1.2em</code>	Vertical gap between systems.
<code>bar-numbers</code>	<code>false str</code>	<code>false</code>	Use <code>systems</code> for later system starts or <code>all</code> for every bar.
<code>first-bar-number</code>	<code>int</code>	<code>1</code>	Positive number assigned to the first bar.

3.9 Indentation and justification

Multi-system scores fill `width` by default: note spacing stretches or compresses so each system's bars fill the line, and adjacent systems keep a similar density rather than alternating cramped and loose lines. There is no fixed bar count per line. A one-system score stays at natural width with the default `ragged-right: auto`.

Use `indent` for the first system and `short-indent` for every later one; both reduce the usable line width while retaining a shared right edge. Set `ragged-last: true` for a natural-width final line.

```
1  #score(  
2    time: "4/4",  
3    width: 36,  
4    indent: 3.5,  
5    ragged-last: true,  
6    bars: (  
7      (notes: "c5:q d e f"),  
8      (notes: "g5:q a b c6"),  
9      (notes: "c6:h g5:h"),  
10     (notes: "f5:q e d c"),  
11     (notes: "b4:h c5:h"),  
12     (notes: "d5:q e f g"),  
13   ),  
14 )
```

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4 Event language

Syntax	Meaning
<code>c4:q</code> or <code>c4q</code>	Quarter note.
<code>ce</code> or <code>c:e</code>	Relative C with an eighth-note duration.
<code>c4:q d e f</code>	Four quarter notes using inherited register and duration.
<code>bb4:e.</code>	Dotted eighth note.
<code>c##5:q</code> / <code>dbb5:q</code>	Double-sharp / double-flat quarter note.
<code>(a4 c e):h (f a c)</code>	Relative chord pitches and inherited duration.
<code>r:q</code>	Quarter rest.
<code>s:q</code>	Invisible quarter spacer: takes time, draws nothing.
<code>_</code>	Automatic rest placeholder.
<code>c2:e g2 @upper e4 g4</code>	Draw the following events on staff <code>upper</code> .
<code>(c3 g3 @upper e4 c5):h</code>	Split chord: one stem across two staves.
<code>~</code>	Tie the previous event to the next event.
<code>/</code>	Break the automatic beam before the next event.
<code>-</code>	Join adjacent flagged events into one beam group.
<code>tuplet 3:2 { c:e d e }</code>	Inline time-scaled music group.
<code>tuplet 3:2[number=never] { c:e d e }</code>	Tuplet without its numeral.
<code>cue { c:e d }</code>	Cue-sized notes that keep their rhythmic value.
<code>acciaccatura { d:e } f:q</code>	Slashed single grace note before F.
<code>tremolo 16 { c:h g:h }</code>	Two-note alternating sixteenth tremolo.

4.1 Duration inheritance

Durations are `w`, `h`, `q`, `e`, `s`, and `t`; append `.` or `..` for dots. An explicit duration becomes the current duration for that staff. Subsequent notes, chords, and ordinary rests may omit it and inherit the complete value, including dots, across barlines. With no preceding value, the default is `q`. The computed-rest placeholder `_` does not change this state, and a written colon must always be followed by a duration. An omitted duration never expands to fill the measure; normal bar-duration validation still reports an incorrect total.

4.2 Pitch spelling and relative octaves

Pitch letters are ASCII case-insensitive. `c4`, `C4`, and mixed-case input have identical pitch and octave semantics; this guide uses lowercase as the compact style, while uppercase remains valid. Compact `ce` and explicit `c:e` both mean an eighth-note C. Whitespace distinguishes that single event from `c e`, which means two notes using the inherited duration. Capitalization never selects an octave.

Pitches accept `#`, `b`, `##`, or `bb`. In lowercase, `bb4` is B-flat 4 and `bbb4` is B-double-flat 4. A single note with an explicit octave anchors later octave-less single notes. Each omitted octave selects the nearest diatonic pitch to the preceding resolved note, so `g4:e a b c` ends on C5 and all four notes are eighths. Accidentals and the active key signature do not affect that register choice. The anchor continues across

bars independently for each staff, and any explicit octave resets it. At the beginning of an unanchored staff, treble, alto, and tenor start in octave 4; bass starts in octave 3.

4.3 Relative chords

Chords follow LilyPond's relative-entry model. The first written chord pitch is resolved from the current staff anchor. Each later chord pitch is resolved from the pitch immediately before it inside the chord. Afterward, the first written chord pitch becomes the external anchor for the next event. For example, (a4 c e):h (f a c) resolves to (A4 C5 E5):h (F4 A4 C5):h. Explicit octaves remain exact overrides for open or otherwise non-nearest voicings.

4.4 Beam boundaries

Standalone / and - are local beam-boundary controls. / splits before the next event; - joins the flagged event on each side even across an automatic metric boundary. A join at an edge, beside a rest, or beside an unflagged duration is an error. For example, this 3/8 measure is grouped 3+3 rather than the automatic 2+2+2 sixteenths, and the next bar demonstrates relative chord voicing. No key argument is needed here: score defaults to C major.

```
1 #score(
2   time: "3/8",
3   beams: true,
4   bars: (
5     (notes: "g4:s a - b / c - d e"),
6     (notes: "(c5 e g):e (d f a) (e g b)"),
7   ),
8 )
```

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4.5 Tuplets

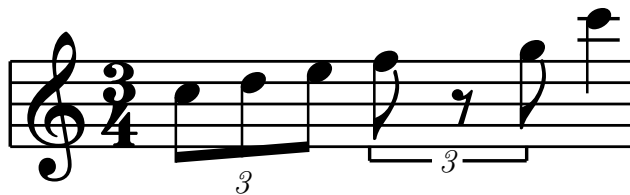
Tuplets are inline music groups: `tuplet 3:2 { c:e d e }` writes three eighths in the time normally occupied by two. Any positive ratio is supported, and groups may nest. The group preserves pitch and duration inheritance, so only its first event needs an explicit octave or duration when the surrounding context already supplies one.

By default the numerator is centered on the group, and a bracket is omitted when one visible beam spans every event in the tuplet; otherwise a bracket is drawn. Written rests keep the bracket. Use optional group controls when the engraving needs an explicit choice: `bracket=always`, `bracket=never`, `side=above`, or `side=below`. `number=never` hides the numeral independently of the bracket, and `number=always` restores the default; combine `number=never` with `bracket=never` to hide the tuplet entirely. Each option may appear once, and unknown values are reported.

```

1 #score(
2   time: "3/4",
3   beams: true,
4   bars: (
5     (
6       notes: "tuplet 3:2 { c5:e d e } tuplet 3:2 { f5:e r:e g5:e } c6:q",
7     ),
8   ),
9 )

```

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4.6 Grace notes

Grace groups stay inline and consume no bar time. `grace { ... }` writes plain small grace notes. `acciaccatura { ... }` adds a slur to the following principal note, as does `appoggiatura { ... }`. For a multi-note group, the slur starts at the first grace event. A single flagged acciaccatura also receives the conventional slash; a multi-note beamed acciaccatura does not. Consecutive flagged grace notes beam as an independent group. Their written durations control flags and beams but do not contribute to the measure total.

```

1 #score(
2   time: "4/4",
3   beams: true,
4   bars: (
5     (notes: "acciaccatura { b4:e } c5:q acciaccatura { d5:s e } f5:q appoggiatura
6       { g5:e } a5:q grace { b5:s c6 } d6:q"),
7   ),
8 )

```

 Typst


A grace group must contain notes or chords and must precede a principal event. Rests, automatic rests, triplets, alternating tremolos, and nested grace groups are rejected inside it so timing remains unambiguous.

4.7 Cue-sized notes

`cue { ... }` draws notes, chords, and written rests at the reduced grace-note size while keeping their full rhythmic value. Unlike grace notes, they count toward the measure total and align with other staves like normal events. Use a cue group for a small but measured figure, such as a triplet upbeat. Cue groups may contain triplets, and triplets may contain cue groups. A triplet whose events are all cue-sized also draws a smaller numeral and bracket.

Cue-sized and normal-sized notes never share a beam: an automatic beam breaks at the group boundary, and a `-` join across it is an error. Fingerings, articulations, dynamics, and other annotations keep their normal size.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    key: "Cm",
7    time: "2/4",
8    beams: true,
9    bars: (
10     (
11       partial: "1/16",
12       upper: "cue { triplet 3:2[number=never] { g4:t d5 d } }",
13       lower: "r:s",
14     ),
15     (
16       upper: "eb5:s f#4 c5 eb d f4 b d",
17       lower: "(g2 c3 g3):e r (g2 c3 g3):e r",
18     ),
19   ),
20 )

```

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Combined with independent voices, a cue group can share a measured upbeat with a normal-sized note. Here the normal G stems upward in voice 1, while the cue-sized triplet stems downward in voice 2 and merges with the coincident G notehead. `side=above` keeps the numeral on the notehead side, clear of the beams.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    key: "Cm",
7    time: "2/4",
8    beams: true,
9    bars: (
10     (
11       partial: "1/16",
12       upper: ("g4:s", "cue { triplet 3:2[side=above] { g4:t d5 d } }"),
13       lower: "r:s",
14     ),
15   ),
16 )

```

 Typst


A cue group must contain at least one note, chord, or written rest. Nested cue groups, grace groups, alternating tremolos, and automatic rests are rejected inside it, as is a beam marker before the group or at its end.

4.8 Tremolos and arpeggios

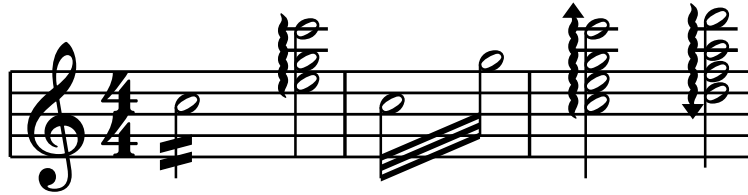
For a repeated single event, append `tremolo=8, 16, 32, or 64` in its annotation block. These values draw one, two, three, or four strokes respectively, independent of the event's written duration. For an alternation, place exactly two equal-duration notes or chords in `tremolo subdivision { ... }`; the connecting strokes replace ordinary flags or beams.

Append `arpeggio`, `arpeggio=up`, or `arpeggio=down` to a chord. The arpeggio wave spans the full chord immediately to its left; the directional forms add an arrow at the appropriate end.

```

1 #score(
2   time: "4/4",
3   bars: (
4     (notes: "c5:h[tremolo=16] (e5 g5 c6):h[arpeggio]"),
5     (notes: "tremolo 16 { c5:h g5:h }"),
6     (notes: "(c5 e5 g5 c6):h[arpeggio=up] (d5 f5 a5 d6):h[arpeggio=down]"),
7   ),
8 )

```

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4.9 Expressive annotations

Annotations follow an event in square brackets. Supported annotations include fingering (`f=4`), staccato, staccatissimo, tenuto, accent, marcato, text directions, ornaments (turns, trills, and mordents), fermatas (`fermata`), breath marks (`breath`), named slurs, pedal spans, hairpins, and dynamics (`dyn=p`, `dyn=pp`, `dyn=mf`, `dyn=sfz`, and the other standard SMuFL combinations).

Every ornament occupies one script slot above the note, so an event may carry at most one of them. The turn family — `turn`, its reverse `inverted-turn`, and the accidental-bearing `chromatic-turn` — accepts a fingered digit through `turn-f=`.

Dynamics, hairpins, slurs, articulations, and fingerings avoid collisions with each other automatically: a hairpin shortens around an adjacent dynamic such as a closing `dyn=ff`, and fingerings rise above stacked articulations rather than overlapping them. Named slurs over beamed passages follow the beam group's stem direction, so a slur goes below when every note it spans stems upward and above when any note stems downward.

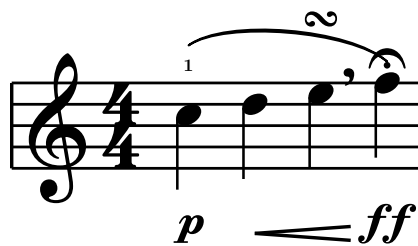
Syntax	Mark
<code>f=4</code>	Fingering.
<code>stacc</code> / <code>staccatissimo</code>	Staccato dot / wedge.
<code>tenuto</code> / <code>accent</code> / <code>marcato</code>	Articulation marks.
<code>turn</code> / <code>inverted-turn</code> / <code>chromatic-turn</code>	Turn ornaments.
<code>trill</code>	Trill (<i>tr</i>).
<code>mordent</code> / <code>inverted-mordent</code>	Lower / upper mordent.
<code>dyn=pp</code> / <code>dyn=sfz</code>	Dynamic marking.
<code>fermata</code> / <code>breath</code>	Fermata / comma breath mark.
<code>s1(... s1)</code>	Named slur.
<code>p1(... p1)</code>	Pedal span.
<code>h1< or h1> ... h1!</code>	Crescendo / diminuendo hairpin.

```

1 #score(
2   clef: "treble",
3   time: "4/4",
4   beams: true,
5   bars: (
6     (notes: "c5:q[f=1 s1( dyn=p) d5:q[h1<] e5:q[turn breath] f5:q[s1) h1! dyn=ff
7       fermata]"),
8   ),

```

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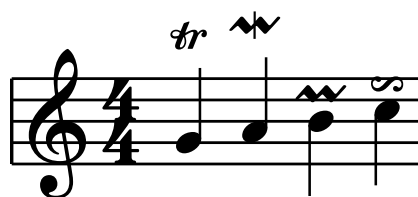
Ornaments share the turn's placement above the note, so trills and mordents stack over articulations and under fingerings just as a turn does.

```

1 #score(
2   clef: "treble",
3   time: "4/4",
4   bars: (
5     (notes: "g4:q[trill] a4:q[mordent] b4:q[inverted-mordent] c5:q[inverted-turn]"),
6   ),
7 )

```

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4.10 Cross-staff notation

Piano music often splits one gesture between the staves: an arpeggio climbs from the bass into the treble, a run crosses the hands, or a chord's lower notes sit on the bass staff while its upper notes sit on the treble. In typed-scores a voice always belongs to its home staff, the staff whose field holds its text, and that is where its rhythm, beams, slurs, dynamics, pedals, and lyrics live. Individual events can be drawn on another staff without leaving the voice.

Staff switches

`@staff` draws every following event of the voice on the named staff, using that staff's clef. The switch holds until the next `@staff` and resets to the home staff at every bar, so each bar's text reads on its own. Write the switch as its own word, separated by spaces. Switches may appear inside tuplets and

cue groups. They may also come just before a grace group, but not inside one or inside an alternating tremolo.

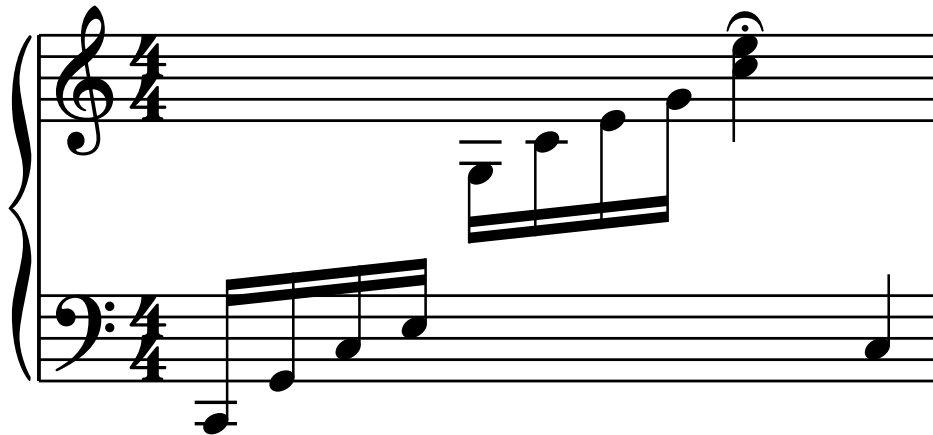
Relative octaves ignore the switch: `@lower g` after `c4` still resolves to the G nearest C4. A note drawn away from its home staff points its stem back toward that staff, so the hand playing it stays visible. Use the spacer `s` where a staff has no notes of its own. It takes time like a rest but draws nothing, and it accepts the same durations, including inherited ones.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    time: "4/4",
7    beams: true,
8    bars: (
9      (
10       upper: "s:h e5:q[fermata] s",
11       lower: "c2:s g2 c3 e3 @upper g3 c4 e4 g4 c5:q @lower c3",
12     ),
13   ),
14 )

```

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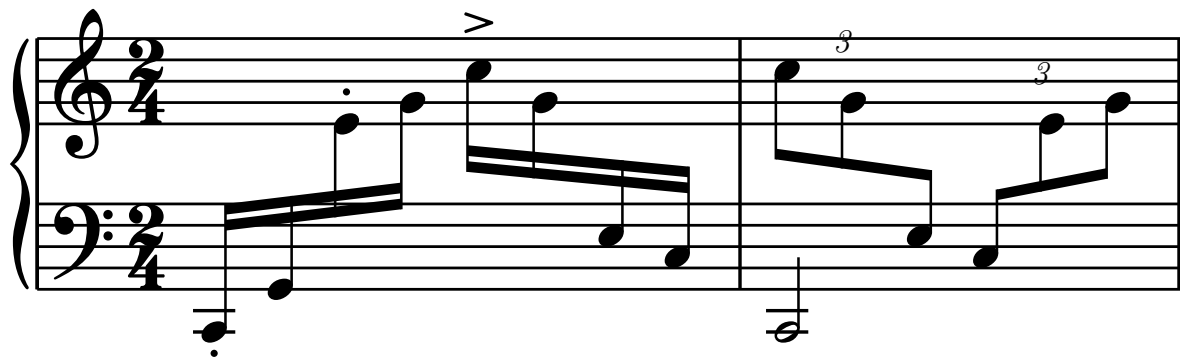
Kneed beams

When one automatic beam group holds notes from two neighboring staves, the beam is kneed: it runs between the staves, stems from the upper staff point down to it, and stems from the lower staff point up. The gap between the staves grows until each stem keeps a comfortable length. Secondary beams stack toward the notes where the group begins. Articulations follow the notehead side of each stem. A voice with forced stem directions keeps them and draws an ordinary beam beside its notes.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    time: "2/4",
7    beams: true,
8    bars: (
9      (
10       upper: "s:h",
11       lower: "c2:s[stacc] g2 @upper e4[stacc] g4 c5[accent] g4 @lower e3 c3",
12     ),
13     (
14       upper: "tuplet 3:2 { c5:e g4 @lower e3 } triplet 3:2 { c3 @upper e4 g }",
15       lower: "c2:h",
16     ),
17   ),
18 )

```

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Split chords

Inside a chord, `@staff` moves the remaining pitches to a neighboring staff, and one stem joins both parts across the gap. The upper staff must hold the chord's upper register. An arpeggio sign on a split chord spans both staves.

```

1  #score(
2    staves: (
3      upper: (clef: "treble"),
4      lower: (clef: "bass"),
5    ),
6    time: "4/4",
7    bars: (
8      (
9        upper: "s:w",
10       lower: "(c3 g3 @upper e4 c5):h (a2 e3 @upper c4 a4):h[arpeggio]",
11      ),
12    ),
13  )

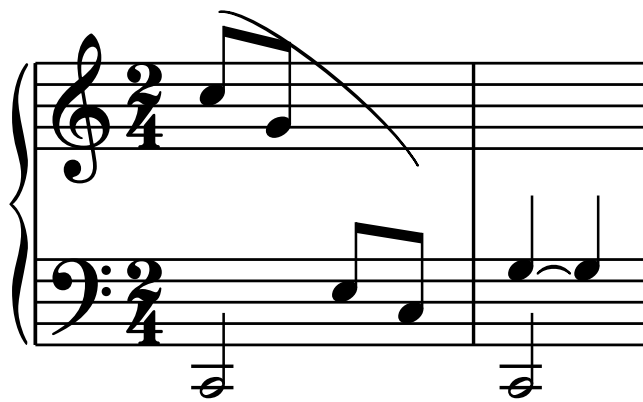
```

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Slurs, ties, and accidentals

Slurs follow the phrase across the staves. A slur whose notes lie on two staves arches above them. A tie needs both notes on the same staff; because switches reset at every bar, a note tied over the barline repeats its `@staff` switch in the next bar. An accidental holds until the end of the bar on the staff that draws it, for every voice drawn on that staff. A note moved to another staff follows that staff's accidentals.

```
1  #score(  
2    staves: (  
3      upper: (clef: "treble"),  
4      lower: (clef: "bass"),  
5    ),  
6    time: "2/4",  
7    beams: true,  
8    bars: (  
9      (upper: "c5:e[s1() g4 @lower e3 c[s1]]", lower: "c2:h"),  
10     (upper: "@lower g3:q ~ g3", lower: "c2:h"),  
11   ),  
12 )
```

 Typst

5 Validation and layout

The package validates every full bar against its active **time** signature and every pickup against **partial**. It also reports unknown staff IDs, missing staff content, malformed or undocumented annotations, impossible automatic rests, invalid slur/pedal/hairpin lifecycles, and tie errors with staff and bar context. A tie must connect immediately adjacent events with the same written pitch or chord; it is split cleanly when the score wraps and suppresses a redundant continuation accidental. Multi-staff onsets share horizontal positions, and bars remain atomic when systems wrap.

Condition	Diagnostic
Malformed pitch or duration	The parser identifies the invalid component.
Wrong full-bar duration	Actual reduced duration and expected meter.
Wrong pickup duration	Actual duration and declared partial value.
Wrong harmony duration	Actual reduced duration and expected meter.
Invalid lyric token count or control	Bar, staff, verse, expected event count, and a concrete correction.
Unknown lyric staff or dangling continuation	The invalid staff reference, hyphen, or melisma source.
Unknown or missing staff ID	The one-based bar and offending field.
Invalid span lifecycle	The slur, pedal, hairpin, or ending that cannot be paired.
Invalid tie or beam join	The adjacent events that cannot legally connect.
Unknown, redundant, or dangling @staff switch	The switch and the staves the score declares.
Split chord out of register or across distant staves	The pitches or staves that one stem cannot join.
Tie across staves or beam across distant staves	The event that must stay on one staff or break its beam.
staff-gap too small for a kneed beam	The minimum gap the beam needs.

5.1 Worked example: Beethoven

The coda of the “Moonlight” Sonata’s Presto agitato sends the right hand’s arpeggios onto the bass staff and back, with cross-staff slurs and kneed sextuplet beams:

L. van Beethoven

5.2 Worked example: Chopin

The canonical release fixture uses the multi-staff form and exercises a pickup, compound beaming, ties, slurs, dynamics, turns, pedal spans, and chromatic accidentals:

The image displays two systems of musical notation, likely for a piano. Both systems are in 12/8 time and use a key signature of two flats (B-flat and E-flat).

The first system begins with a piano (*p*) dynamic. The right hand features a melodic line with various ornaments (accents, slurs, and fingerings) and a final triplet. The left hand provides a harmonic accompaniment with chords and eighth-note patterns.

The second system begins with a forte (*f*) dynamic. The right hand continues the melodic development with more complex ornaments and a final triplet. The left hand maintains the harmonic accompaniment, concluding the system with a double bar line.

6 Release showcase

The repository’s `examples/` directory contains six reusable, source-checked fixtures and a compiled `showcase.pdf`:

- Chopin, Nocturne Op. 9 No. 2 — piano, pickup and measures 1–4;
- Mozart, Eine kleine Nachtmusik K. 525/I — four-staff strings, measures 1–4;
- Bach, Cello Suite No. 1 BWV 1007/I — solo cello, measures 1–4;
- Beethoven, the four-bar Ode to Joy theme — E-flat alto saxophone at written pitch;
- Beethoven, Für Elise WoO 59 — piano, pickup and measures 1–4; and
- Beethoven, Piano Sonata Op. 27 No. 2/III — piano, cross-staff coda, measures 178–185.

The saxophone fixture is an explicit transposition: its B-major notation sounds in Beethoven’s D major. The others retain their reference score’s written pitches. This compact quartet excerpt demonstrates a bracketed ensemble:

W. A. Mozart

Allegro

The image displays a musical score for a string quartet, specifically an excerpt from Wolfgang Amadeus Mozart's 'Eine kleine Nachtmusik' (K. 525/I). The score is written for four staves: Violin I, Violin II, Viola, and Violoncello. The key signature is D major (two sharps) and the time signature is 4/4. The tempo marking is 'Allegro'. The score is presented in a compact, bracketed format, showing measures 1 through 12. The first system (measures 1-4) begins with a forte (f) dynamic. The second system (measures 5-8) shows the Violin I part playing a melodic line, while the other instruments provide harmonic support. The third system (measures 9-12) continues the ensemble. The score is presented in a compact, bracketed format, showing measures 1 through 12.

The single-staff case needs no staff declaration, even for a transposing instrument whose written pitch has been prepared by the author:

Allegro assai

L. van Beethoven

A musical score snippet in treble clef, key of D major (two sharps), and 4/4 time. The notation consists of a single staff with a series of eighth notes and a final half note. The notes are: D4, E4, F#4, G4, A4, B4, C#5, D5, C#5, B4, A4, G4, F#4, E4, D4. The first four notes are beamed together. The fifth and sixth notes are beamed together. The seventh note is a half note. The eighth note is a quarter note. The ninth and tenth notes are beamed together. The eleventh and twelfth notes are beamed together. The thirteenth note is a half note. The fourteenth note is a quarter note. The fifteenth note is a half note. The dynamic marking *p* is placed below the first note.

7 Limitations

7.1 Current scope

- One to four independent rhythmic voices per staff are supported; the count for a staff is fixed across its bars.
- Cross-staff beams and split chords join two neighboring staves; stems cannot be flipped by hand, and ties cannot cross staves.
- Grace groups currently exclude rests, tuplets, and nested ornamental groups.
- Arpeggio signs span one chord; a cross-staff arpeggio is written on a split chord.
- Pedals and hairpins are event-anchored and do not split automatically at a system break.
- Dense editorial markings or lyric verses may require **staff-gap**, **note-spacing**, **lyric-gap**, **verse-gap**, or **scale** adjustments.

7.2 Grammar at a glance

Text	Meaning
<code>c4:q / c4q</code>	Note with an explicit octave and duration.
<code>c4:q d e f</code>	Relative pitches inheriting the quarter duration.
<code>(a4 c e):h</code>	Relative chord.
<code>r:q / _</code>	Written rest / computed remainder rest.
<code>s:q</code>	Invisible spacer that fills time.
<code>@lower</code>	Draw the following events on another staff until the bar ends.
<code>(c3 @upper e4):h</code>	Split chord across two neighboring staves.
<code>~</code>	Tie the preceding event.
<code>/ / -</code>	Break / force a local beam connection.
<code>tuplet 3:2 { c:e d e }</code>	Three written eighths in the time of two.
<code>tuplet 3:2[number=never] { c:e d e }</code>	Tuplet with its numeral hidden.
<code>cue { c:e d }</code>	Small notes that still occupy bar time.
<code>acciaccatura { d:e } f:q</code>	Slashed single grace note resolving to F.
<code>tremolo 16 { c:h g:h }</code>	Two-note alternating tremolo.
<code>(c e g):h[arpeggio=up]</code>	Upward arpeggio over a chord.
<code>lyrics: "Twin -- kle _ _"</code>	Syllables with a hyphen, melisma extender, and skip.
<code>[stacc dyn=pp]</code>	One annotation block with multiple marks.
<code>barline: (left: "repeat-start")</code>	Repeat sign on a bar edge.
<code>barline: (right: "final")</code>	Final barline on the right edge.
<code>clef: "bass"</code>	Persistent single-staff clef change at a bar boundary.
<code>rehearsal: "A" / navigation: "segno"</code>	Boundary rehearsal / navigation mark.
<code>ending: (label: "Final", start: true)</code>	Literal volta label.

7.3 License and credits

Package code is released under the MIT License. Bundled Bravura glyphs are released under the SIL Open Font License 1.1. `typed-scores` uses CeTZ for drawing and Bravura’s SMuFL metadata for music-glyph geometry.

The Chopin, Mozart, Bach, and Beethoven compositions used by the release fixtures are public domain. Their reusable Typst sources live in the `examples/` directory and are shared by the guide, showcase, and visual regression suite.